Aniruddh Herle

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Research Profile

Cosmologist working at the interface of large-scale structure, intrinsic alignments, and hydrodynamical simulations. My research uses cosmological simulations (FLAMINGO, COLIBRE), weak lensing statistics, and statistical methodology to understand the key systematic in next-generation weak lensing analysis. I develop robust two-point statistical estimators and modelling strategies that will allow precision cosmology with *Euclid*, LSST and Roman.

Current Position

2023–2027 **PhD Candidate in Cosmology**, Leiden Observatory, The Netherlands
Advisors: Prof. Henk Hoekstra (Leiden University), Dr. Elisa Chisari (Utrecht University)
Title: Intrinsic alignments in hydrodynamical simulations. Research focus: intrinsic alignments, halo shapes, weak lensing statistics, hydrodynamical simulations, galaxy–halo connection.
Expected graduation: Sept 2027

Education

2021–2023 **MSc Astrophysics**, Ludwig Maximilian University of Munich, Germany
Grade: 1.48 (Distinction in German system)
Thesis: Selection functions of strong lens finding neural networks (publication)
Advisors: Dr. Simona Vegetti (Max Planck Institute for Astrophysics), Prof. Dr. Daniel Gruen (University Observatory of Munich)

2017–2021 **B.E. Electrical and Electronics Engineering**, M. S. Ramaiah Institute of Technology, India
CGPA: 9.38/10 (Distinction)

First Author Publications

2026 **A. Herle**, N. E. Chisari, H. Hoekstra, D. Neumann
"Assembly bias and the redshift evolution of intrinsic alignments for LRGs"; *Astronomy and Astrophysics*.
Analysis of redshift evolution of the alignment signal in hydrodynamical simulations, with recommendations for observational modelling for *Euclid*, LSST and Roman, as well as the first analysis of assembly bias on IA.

2026 **A. Herle**, N. E. Chisari, H. Hoekstra et al.
"Intrinsic alignments in the FLAMINGO simulations with two-point statistics"; *Astronomy and Astrophysics*.
Comprehensive analysis of intrinsic alignment modelling in next-generation hydrodynamical simulations with the introduction of the TATT-M model, which will be the fiducial alignment model for *Euclid*.

2025 **A. Herle**, N. E. Chisari, H. Hoekstra et al.
"Unbiased estimates of the shapes of haloes using the positions of satellite galaxies"; *Astronomy and Astrophysics*.
Introduced a statistically unbiased halo shape estimator robust to satellite sampling limitations.

2023 **A. Herle**, C. M. O'Riordan, S. Vegetti
"Selection functions of strong lens finding neural networks"; *Monthly Notices of the Royal Astronomical Society*.
Quantified detection biases in machine-learning-based strong lens searches.

2020–2022 A total of 5 First-author publications and several co-authored publications on machine learning applications to quasar detection and Li-ion battery modelling. Published in IEEE conferences and the *International Journal of Energy Research*. Full list available on Google Scholar and ADS.

Close to submission

2026 **A. Herle**, P. Vielzeuf, N. E. Chisari, H. Hoekstra et al.
"Alignments in the emptiest regions of the universe: IA of galaxies around voids in the FLAMINGO simulations" *(in prep.)*

Supervision

2026 Rahul Shinde – Robust TATT-M modelling with the FLAMINGO simulations (ongoing).
2026 Antonis Sapounas – Reducing false positives in strong lens finding with *Euclid* DR1 (ongoing).
2026 Kostas Akalestos – Dark matter halo shape recovery in mock *Euclid* observations (ongoing).
2025 Lea Krarup – Substructure detection in weak lensing maps of galaxy clusters (successfully defended).

Teaching Experience

2024–2027 Teaching Assistant, "Large Scale Structure and Galaxy Formation"; Instructor: Prof. Dr. Joop Schaye
Led weekly tutorial sessions for ∼40 MSc students each year; responsible for marking and teaching problem sets, holding office hours, and giving feedback on written assignments.

Community Initiatives

2026 **Hack lead**, UNBIAS echo-IA Workshop: Moderated a working group discussion to define community requirements for next-generation hydrodynamical simulations for intrinsic alignment science
2025 **Hack lead**, CAROLINA echo-IA Meeting: Organised and led a community code-comparison effort to validate consistency across IA modelling codes; developed a public package (IATheory)

Service, Outreach and Leadership

2026 **Local Organising Committee**, UNBIAS echo-IA Annual Meeting
Conference planning, session moderation and website design.
2023–2027 **Co-organiser, Astronomy on Tap Leiden**
Monthly public outreach events; ∼70 attendees per event over four years.
2024–2025 **Core member, Social Committee**, Leiden Observatory.
2023–2026 **Expert Referee** — ApJ, *Euclid* internal review, RASTI.

Contributed Talks at international conferences

2026 Princeton Cosmology talk, Princeton, USA (*upcoming*)
KIPAC Tea talk, SLAC, Stanford, USA (*upcoming*)
FLAT talk, University College London, London, UK
CMB-LSS talk, Kavli Institute for Cosmology, Cambridge, UK
COSMO-26, Leiden University, Leiden, the Netherlands
UNBIAS echo-IA workshop, Utrecht University, Utrecht, the Netherlands
Intrinsic alignment workshop, IAP, Paris, France
2025 NOVA Network 1 Cosmology Meeting, Leiden University, Leiden, the Netherlands
Dark Universe Science Collaboration Meeting, University of Groningen, Groningen, the Netherlands
CAROLINA echo-IA workshop, Durham, Duke University, USA
Lunch Talk, Observatory of Munich, Munich, Germany
Lunch Talk, University of Strasbourg, Strasbourg, France
2024 MOCK Barcelona, INAF, Barcelona, Spain
ICTP Summer School, ICTP, Trieste, Italy
2023 Machine Learning in Astronomy, Paris Observatory, Paris, France

Technical Expertise

• Hydrodynamical simulations: FLAMINGO, COLIBRE • Advanced statistical analysis of large-scale structure and intrinsic alignments • Two-point statistics and correlation function estimators • High-performance computing (SLURM clusters, GPU acceleration) • Python scientific stack with years of experience (NumPy, SciPy, Astropy, PyTorch) • Machine learning for astrophysical data analysis